SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I **SAVITHA R 192421368** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **SAVITHA R**

Short Answer Test

1. **A classification model produces the following confusion matrix values: True Positives = 120, True Negatives = 200, False Positives = 30, and False Negatives = 50. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on these results, evaluate whether the model is suitable for applications where false negatives are critical.**

Given:

True Positive (TP) = 120

True Negative (TN) = 200

False Positive (FP) = 30

False Negative (FN) = 50

Step 1: Calculate Total Cases

N=TP+TN+FP+FN

N=120+200+30+50=400

Step 2: Accuracy

Accuracy=(TP+TN)/Total=(120+200)/400=320/400=0.80

Accuracy = 80%

Step 3: Precision

Precision=TP/(TP+FP)=120/(120+30)=120/150=0.80

Precision = 80%

Step 4: Recall (Sensitivity)

Recall=TP/(TP+FN)=120/(120+50)=120/170=0.7059

Recall ≈ 70.59%

Step 5: Specificity

Specificity=TN/(TN+FP)=200/(200+30)=200/230=0.8696
Specificity ≈ 86.96%

Step 6: F1-Score

F1=2PR/(P+R)

=2(0.80)(0.7059)/(0.80+0.7059)=0.75

F1-Score ≈ 75%

Conclusion

The model performs reasonably well, but the recall of 70.59% means several positive cases are missed. Since false negatives are critical, the model should be improved before deployment.

2. **A Genetic Algorithm is used to maximize the function f(x) = 2x + 3 for an initial population of values {1, 3, 5, 7}. Compute the fitness of each individual and select the top two candidates. Perform a single-point crossover to generate offspring and explain how mutation can help avoid premature convergence.**

Given

Fitness Function:

$f(x)=2x+3$

Population = {1, 3, 5, 7}

Step 1: Calculate Fitness

Individual	Fitness
1	5
3	9
5	13
7	17

Step 2: Select Best Candidates

Best two individuals:

7 → 17
5 → 13

Step 3: Single Point Crossover

Binary Representation:

7 = 111

5 = 101

After crossover (after second bit):

Parent 1 = 11|1

Parent 2 = 10|1

Offspring:

111 = 7
101 = 5

Step 4: Mutation

Example:

111 → 110

7 → 6

Mutation introduces diversity, prevents premature convergence, and helps discover better solutions.

Final Answer

Fitness = {5, 9, 13, 17}

Best Parents = 7 and 5

Mutation improves diversity and search efficiency.

3**. During training of a neural network, the training loss decreases continuously, while validation loss decreases initially and then starts increasing after several epochs. Identify the issue occurring in the model. Explain the reason behind this behavior and suggest two technique .**

Observation

- Training loss decreases continuously.
- Validation loss decreases initially but later increases.

Issue

The model is experiencing **Overfitting**.

Reason

The neural network begins memorizing the training data instead of learning general patterns. This reduces its performance on unseen validation data.

Techniques to Improve Performance

1. **Early Stopping**
   - Stop training when validation loss begins increasing.
2. **Regularization (L2)**
   - Penalizes large weights and reduces overfitting.

Conclusion

The model should stop training at the point where validation loss is minimum. Early stopping and regularization improve generalization.

4.**A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.**

Given

- Advertisement Exposure = 2
- Income Level = 1
- Weight_1 = 0.5
- Weight_2 = 0.3
- Bias = –0.4

Step 1: Calculate Net Input

$$\square\square\square\square\square\square = (0.5)(2) + (0.3)(1) - 0.4$$
$$= 1 + 0.3 - 0.4 = 0.9$$

Step 2: Step Activation Function

Since

$$0.9 \geq 0$$

Output = **1**

Step 3: Decision

Assume:

- Output 1 → Purchase
- Output 0 → No Purchase

Since output = **1**
Customer will purchase the product.

Final Answer

- Net Input = **0.9**
- Output = **1**
- **Purchase = Yes**

5.**In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration.**

Given

- Initial Weight = 0.6
- Learning Rate = 0.15
- Input = 2.5
- Actual Output = 0.7
- Target Output = 1

Step 1: Error

$$Error = 1 - 0.7 = 0.3$$

Step 2: Weight Update

$$\begin{aligned}\Delta w &= \eta(t - y)x \\ &= 0.15(0.3)(2.5) \\ &= 0.1125\end{aligned}$$

Step 3: New Weight

$$\begin{aligned}w_{new} &= 0.6 + 0.1125 \\ &= 0.7125\end{aligned}$$

Final Answer

- Error = **0.3**
- Weight Update = **0.1125**
- New Weight = **0.7125**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand